

## 5.2 BLIP and Captioning

CoCa adds a decoder. BLIP also filters the web: ITC, ITM, and a language-model loss.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

**BLIP** (Li et al., 2022) is a vision–language model that **matches** and **writes**. CLIP scores pairs. CS231N 2025 L16’s CoCa slide is the sibling idea: add a **decoder**. BLIP does that, and it also **cleans** the noisy web pairs CLIP trained on.

**First time this method appears.** **BLIP** is CLIP-style scoring **plus** a decoder that writes text, with a filter on noisy web pairs.

**What.** Three losses: ITC (contrastive), ITM (matched vs unmatched pair), LM (caption tokens).

Bootstrap: generate captions, filter, train. **Why.** CLIP cannot decode a sentence. Naive captioners

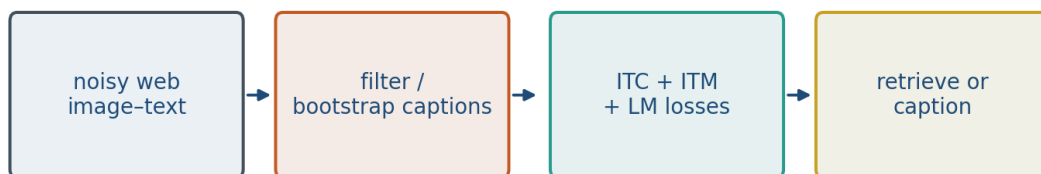
train on noisy alt-text. BLIP tries to clean the web and add generation. **Architecture.** Image encoder, text encoder, and a decoder with **cross-attention** into image tokens (queries from text, keys/values from vision). **How.** ITC aligns. ITM is a binary match head. LM is next-token on the caption.

Generation uses the decoder; retrieval can use ITC embeddings. **Formula.** ITC as InfoNCE; ITM as logistic on a CLS; LM as  $\sum_t -\log p(w_t | w_{<t}, \text{image})$ . **Tradeoffs.** + Captions and retrieval in one family. – Hallucinated captions; filter is extra machinery; heavier than frozen CLIP.

### 1. Bootstrap the data

Web alt-text is messy. BLIP’s captioner generates synthetic captions; a filter keeps the ones that still match the image (ITM: image–text matching). The cleaned pairs retrain the model. That loop is the “bootstrapped pretraining” in the syllabus.

BLIP: clean the pairs, then learn to match and to write



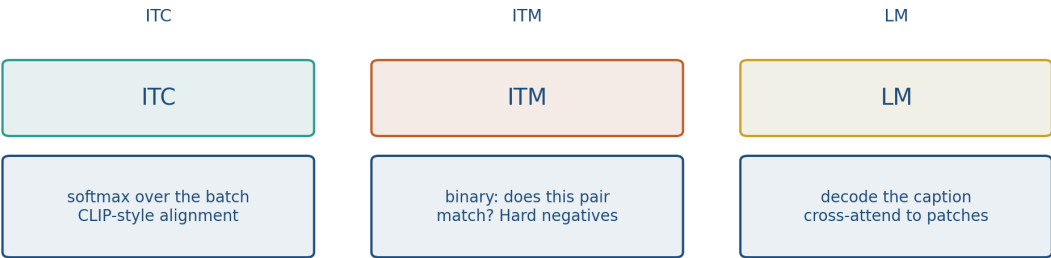
Alt-text like `IMG_4032` or a product SKU is a bad caption. Generating a sentence and then asking “does this still match?” is the filter. You will not run that loop in lab; you will remember it when your project scrapes the web.

CoCa (Yu et al.; the CS231N generation slide) adds a captioning loss on top of contrastive training. It does **not** advertise the bootstrap filter. If your story is “noisy web text,” you need BLIP’s ITM filter, not only a decoder.

## 2. Three losses, one backbone

| Loss                   | Job                                                           |
|------------------------|---------------------------------------------------------------|
| ITC (contrastive)      | CLIP-style alignment                                          |
| ITM (matching)         | Binary: does this caption belong to this image?               |
| LM (language modeling) | Decode the caption, causal on text, cross-attend to the image |

BLIP keeps three losses. CoCa (CS231N) adds a decoder; BLIP also filters the web.



The image tower is a ViT. The text side is a transformer that can encode or decode depending on the head. **BLIP-2** later freezes a strong image encoder and trains a thin **Q-Former**; treat that as “same idea, cheaper.” LLaVA (CS231N’s next family) is a frozen CLIP tower plus an LLM. Not this hour.

ITC needs a batch of negatives. ITM can use a **hard** negative (a caption that almost matches). LM is why BLIP can caption at all.

Whisper, which you will watch as an analog, is an encoder–decoder for **speech** → **text**. BLIP is encoder–decoder for **image** → **text**, plus ITC and ITM.

## 3. Captioning versus retrieval

Retrieval (CLIP) returns an existing string from a gallery. Captioning (BLIP) **generates** a new string. Evaluation: CIDEr / CLIP-score / human, not just recall@k. Hallucinated objects are the usual failure (“a red bike” when there is none).

CLIP-score as a caption metric is circular if your captioner was trained to match CLIP. Prefer at least one human or task-specific check. CIDEr rewards n-gram overlap with reference captions; it punishes valid paraphrases. Report both and show two example captions, one good and one hallucinated.

## 4. Teaching this note

**30–40 minutes.** CLIP retrieves vs BLIP writes, then the three-loss table, then one numeric ITM vs ITC contrast. Reading: the BLIP paper (ITC+ITM+LM). Play the Whisper video as the **encoder–decoder analog** for captioning (**0:00–8:00**). Say explicitly: Whisper does not add ITC+ITM; **BLIP does**.

Minute plan: 8 min retrieve vs generate; 10 min ITC/ITM/LM table; 8 min ITM numbers; 12 min Whisper clip plus the one-sentence BLIP add-on.

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## 5. Worked example

One image, two captions: true “a red mug on a desk,” false “a blue bicycle.”

ITC ( $N = 2$ ): softmax over the batch; the false caption is one negative among many in a real batch.

ITM: a classifier on the fused `[CLS]` with labels 1 and 0. Suppose logits  $s_+ = 2$ ,  $s_- = -1$ .

$$\sigma(s) = \frac{1}{1 + e^{-s}}, \quad \sigma(2) \approx 0.88, \quad \sigma(-1) \approx 0.27.$$

Binary cross-entropy: matched term  $-\log \sigma(2) \approx 0.13$ ; mismatched term  $-\log(1 - \sigma(-1)) \approx 0.31$ .

ITM is not a batch softmax. One pair, a yes/no score.

| ITM logit         | sigmoid                   | binary CE term                   |
|-------------------|---------------------------|----------------------------------|
| matched $s = 2$   | $\sigma(2) \approx 0.88$  | $-\log \sigma \approx 0.13$      |
| mismatch $s = -1$ | $\sigma(-1) \approx 0.27$ | $-\log(1 - \sigma) \approx 0.31$ |

LM: decode `a , red , mug , ...` with cross-attention into ViT patch tokens. Loss is next-token, like GPT, but the “prefix” includes the image.

If alt-text was `DSC0001.jpg`, LM on that string teaches garbage. Filter first, then LM.

ITC wants a **batch**. ITM can score one pair. LM wants a **token sequence**. If your project only retrieves, stop at ITC (or CLIP). If it must write, you need LM. If web text is junk, you need the filter (ITM).

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## 6. Where students get stuck

- Using recall@k as the only captioning metric.
- Thinking ITM and ITC are the same softmax.

- Skipping the filter and training LM on raw alt-text.
  - Calling CoCa and BLIP the same paper. Shared idea: a decoder. BLIP's extra: bootstrap + ITM.
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## 7. Video

Keep the **BLIP paper** as the reading (Li et al., 2022). For a lecture clip, watch [OpenAI's Whisper Model Explained](#) as an **encoder-decoder analog for captioning**: audio encoder, text decoder. Pause on that cross-attention picture.

Then say: **BLIP adds ITC + ITM + LM** on image-text pairs, and bootstraps noisy web captions. Whisper transcribes; BLIP captions and matches.

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## 8. Practice

1. Why filter generated captions instead of trusting every alt-text on the web?
2. ITM vs ITC: one is a binary matcher, one is a softmax over the batch. When would ITM catch a pair ITC misses?
3. Name one metric you would report in a captioning project that CLIP retrieval would not need.
4. ITM logits  $s = 0$  vs  $s = 2$  for a matched pair. Compute  $\sigma(s)$  in both cases. Write the binary-CE term  $-\log\sigma(s)$ .
5. A caption of 6 tokens (including `EOS`). LM loss is mean token NLL. If five tokens have probability 0.5 and `EOS` has probability 0.25, what is the mean NLL? (Use log base  $e$ :  $-\log 0.5 \approx 0.69$ ,  $-\log 0.25 \approx 1.39$ .)